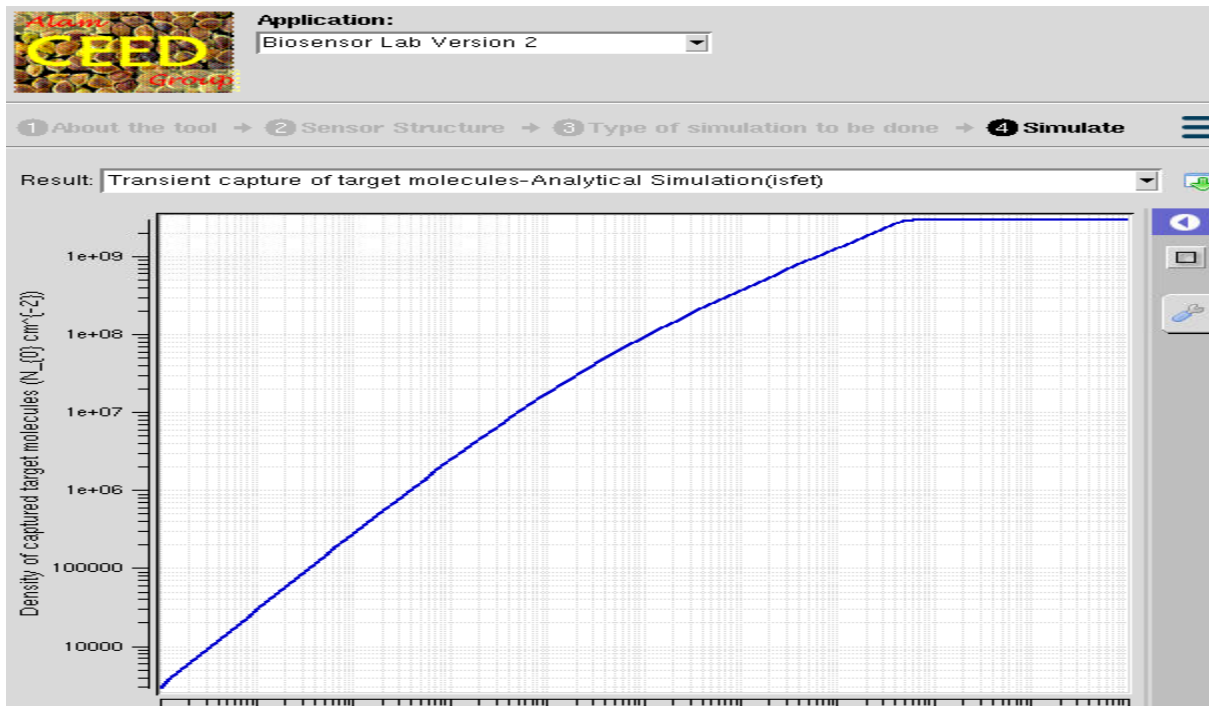


Exp. 10. Biosensor and Bio-FET characteristics using NanoHub

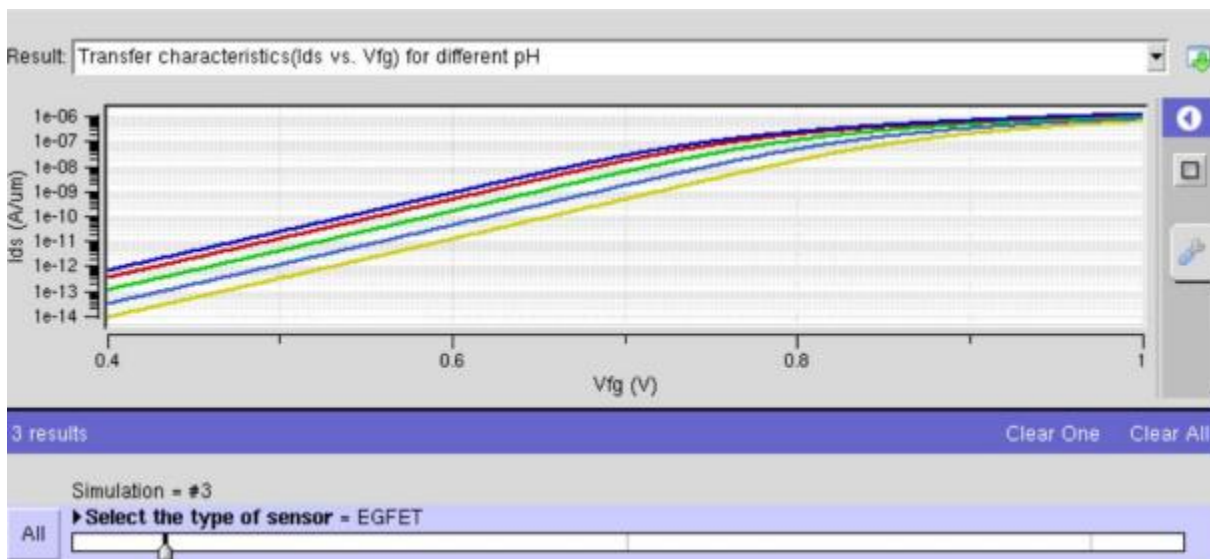
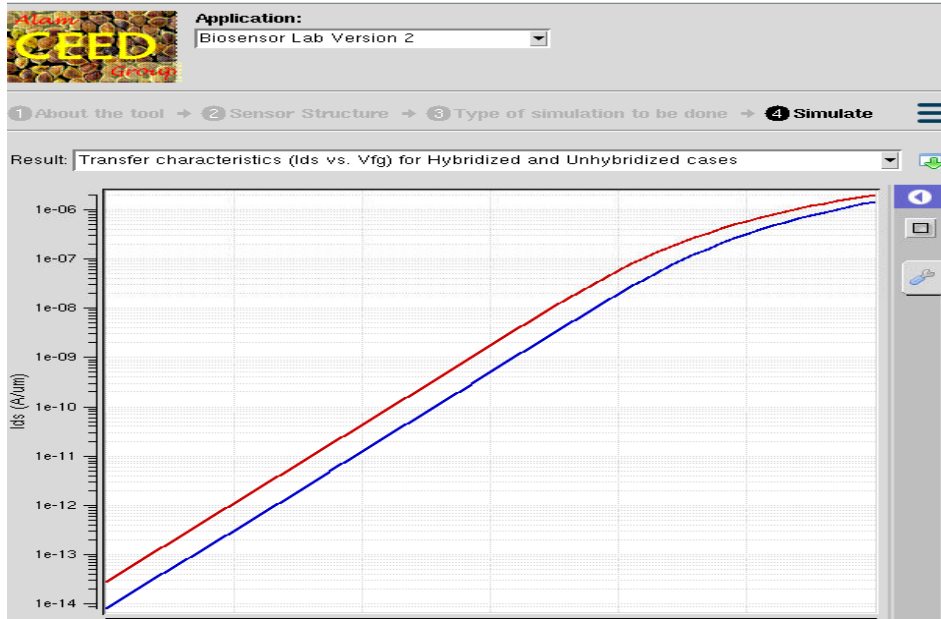
Aim:

1. To simulate the threshold voltage of a BioFET-based biosensor as a function of electrolyte pH (4 to 8).
2. To analyze the transfer characteristics (I_{ds} vs. V_{fg}) of the BioFET for varying pH levels and interpret the sensor's performance.



Observation

1. The drain current (I_{ds}) increases with the front gate voltage (V_{fg}), showing the normal transfer characteristics of the BioFET.
2. Different pH values produce shifted I_{ds} – V_{fg} curves, indicating that the drain current is influenced by the electrolyte pH through changes in the surface charge.
3. Higher pH values result in a noticeable shift in the transfer characteristics, demonstrating the high sensitivity, good response, and reliable sensing performance of the BioFET for pH detection.



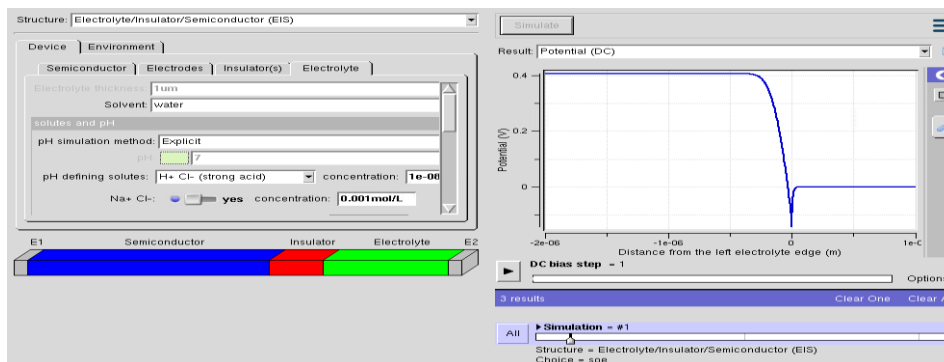
Conclusion:

1. The BioFET sensor demonstrates linear sensitivity to pH changes, with a near-theoretical response of 59 mV/pH⁵⁹ .
2. The transfer characteristics (I_{ds} vs. V_{fg}) effectively shift with varying pH, confirming the sensor's ability to detect electrolyte variations.
3. The EGFET configuration is validated as a reliable model for BioFET simulations, offering insights into potential optimizations for practical .

Exp. 11. Calculate the Potential of a pH SENSOR using NanoHub

Aim: To obtain the Potential of a sensor by varying the pH value of an electrolyte and insulator

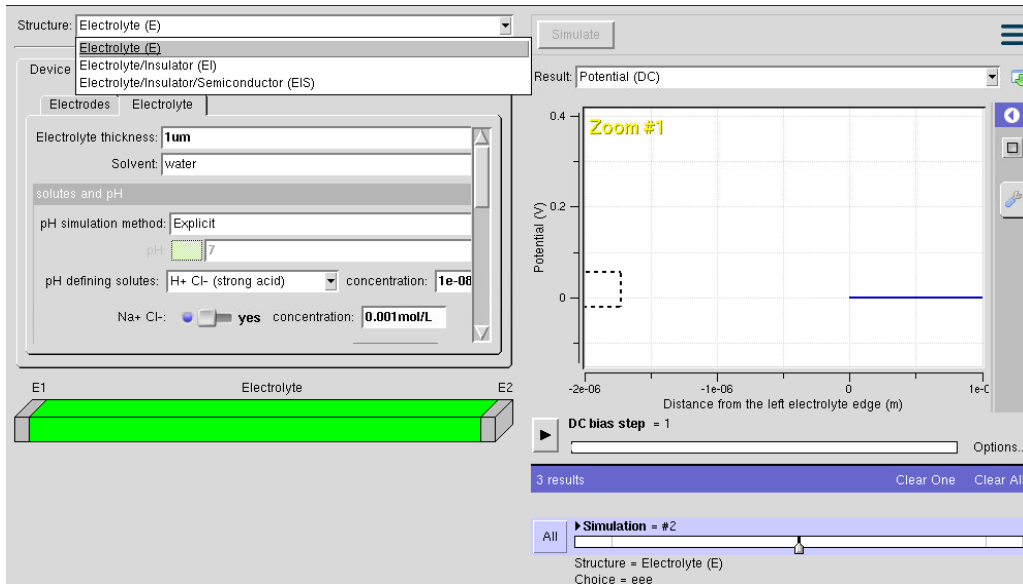
1.Semiconductor/insulator/electrolyte



Observation:

- The EIS structure consists of semiconductor, insulator, and electrolyte layers.
- The electrolyte thickness is 1 μm and water is used as the solvent.
- The simulation is performed at pH 7 with 0.001 mol/L NaCl.
- The potential is approximately 0.4 V in the semiconductor region.
- A sharp potential variation is observed near the semiconductor/insulator/electrolyte interface.
- The potential becomes nearly constant in the electrolyte region

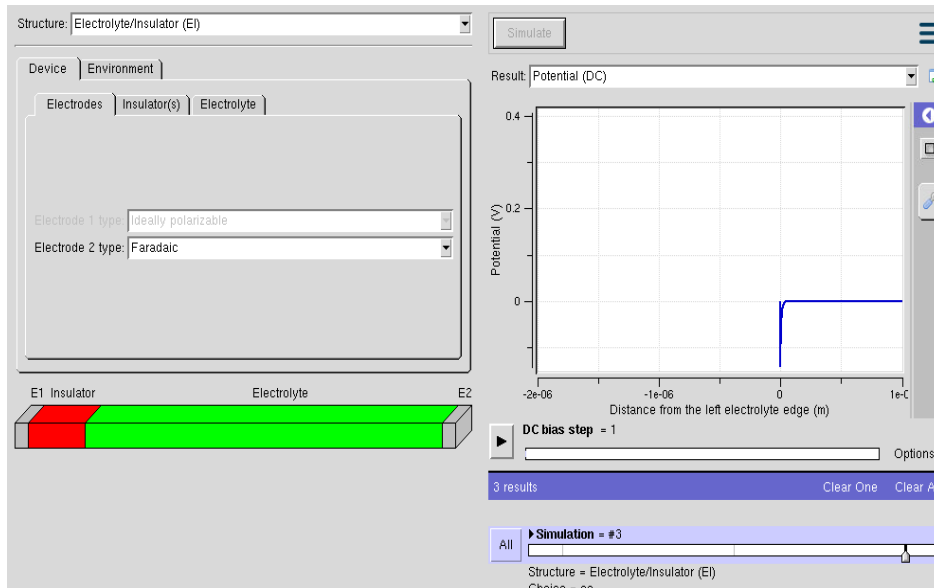
2.Electrolyte



Observation

- The simulation represents an Electrolyte (E) structure.
- The electrolyte thickness is 1 μm , and water is used as the solvent.
- The electrolyte is maintained at pH = 7, indicating a neutral solution.
- The NaCl concentration is 0.001 mol/L.
- The potential is plotted as a function of distance from the left electrolyte edge.
- The potential in the electrolyte region is approximately 0 V and remains almost constant.
- No significant potential variation is observed within the electrolyte because there is no semiconductor or insulator layer in this structure.

3. Insulator



Observation

1. The simulation represents an Electrolyte/Insulator (EI) structure consisting of an insulator and an electrolyte region.
2. The graph shows the DC potential distribution as a function of distance from the left electrolyte edge.
3. A significant potential drop occurs near the insulator–electrolyte interface, indicating a change in electric potential at the interface.
4. In the electrolyte region, the potential becomes nearly constant at approximately 0 V.
5. The result shows that the insulator influences the potential distribution, which is important for understanding electrolyte–insulator interfaces in sensor and Bio-FET applications.

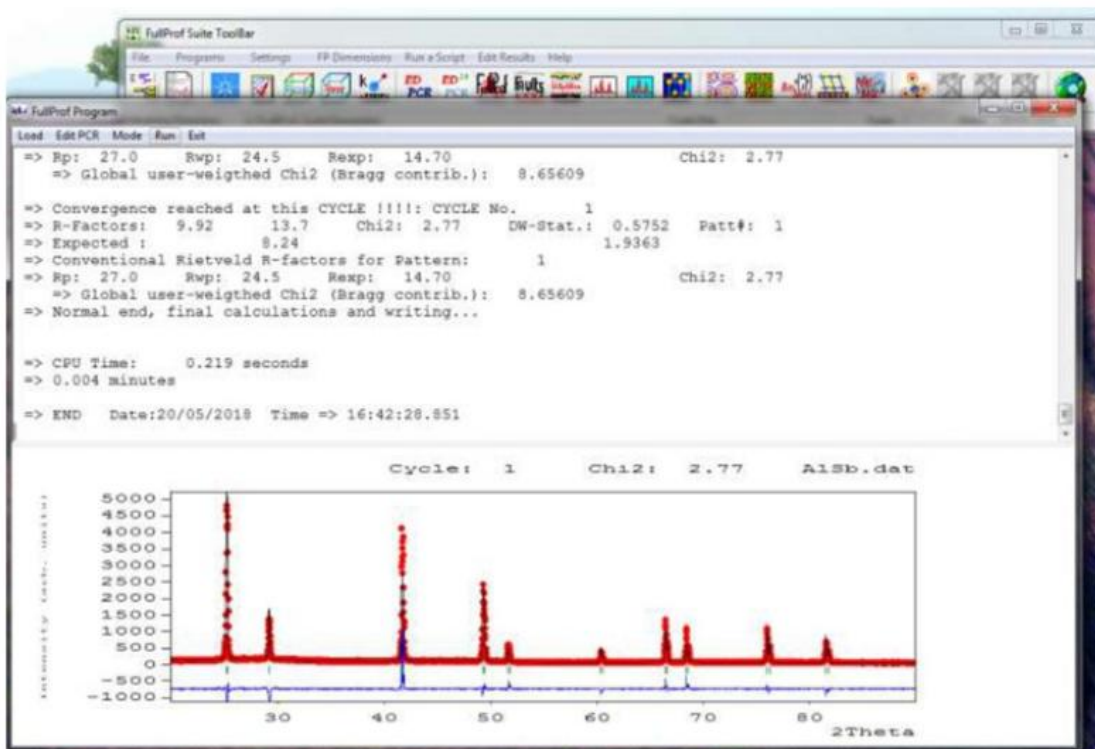
Conclusion

The simulation successfully analyzed the potential distribution in Electrolyte (E), Electrolyte/Insulator (EI), and Electrolyte/Insulator/Semiconductor (EIS) structures. The electrolyte structure showed an almost constant potential, while the EI structure showed a potential variation near the insulator–electrolyte interface. The EIS structure exhibited a more significant potential drop near the semiconductor–insulator–electrolyte interface. These results demonstrate that the presence of insulating and semiconductor layers strongly influences the electrostatic potential at the interface. Overall, the simulation helps in understanding the electrical behavior of electrolyte-based sensors, ISFETs, and Bio-FETs.

Exp. 12. Rietveld refinement & lattice parameter determination (single-phase material) using Full Prof software

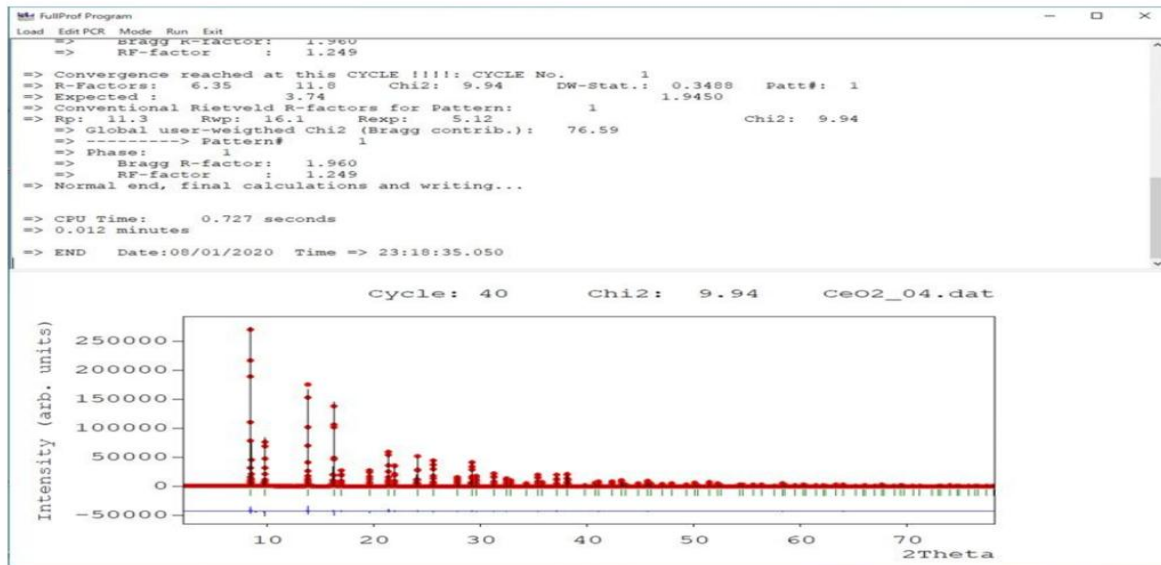
Aim: To perform Rietveld refinement of X-ray diffraction (XRD) data and determine the lattice parameters of crystalline materials using Full Prof software.

Output:



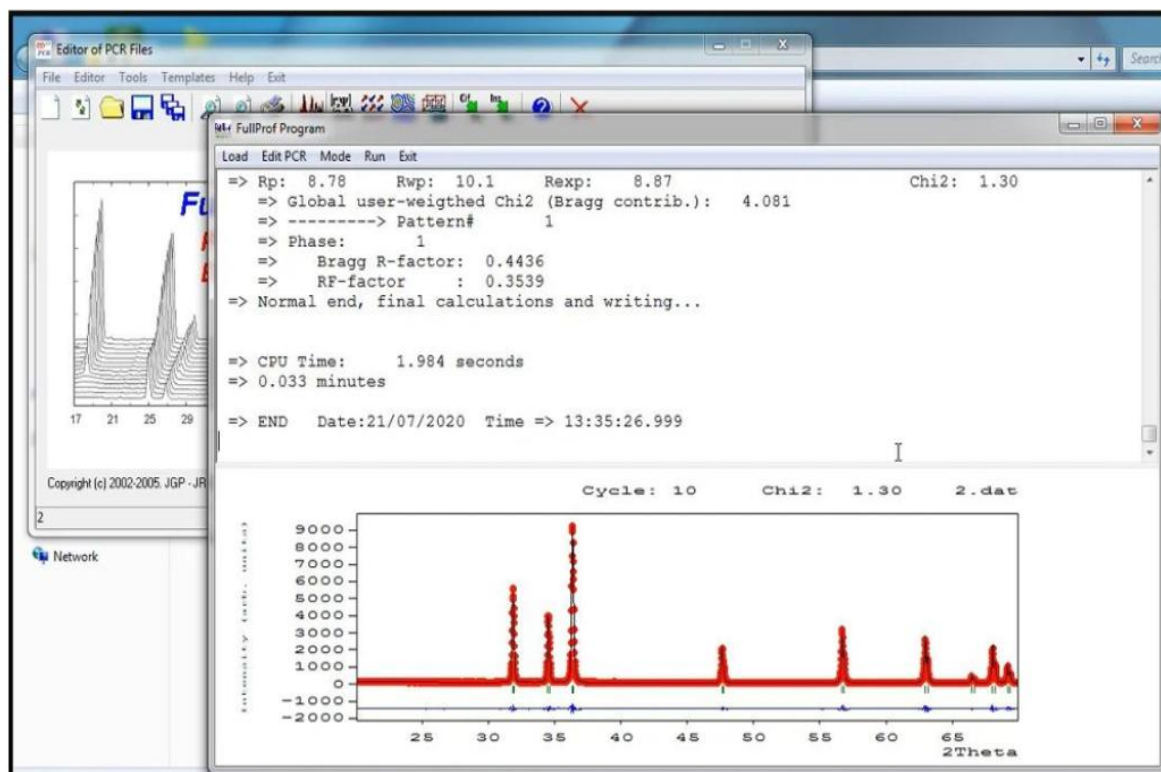
This graph displays a Rietveld refinement of an X-ray diffraction pattern for AlSb, plotting X-ray intensity against the 2-Theta angle. The strong overlap of the experimental data (red) and the low residual baseline (blue) yields an excellent goodness-of-fit ($\chi^2 = 2.77$), confirming an accurate crystal structure model.

Case Study 1: Perform Rietveld refinement for a metal oxide material (CeO_2) using the pseudo Voigt peak profile. Analyze the Rwp and χ^2 values.



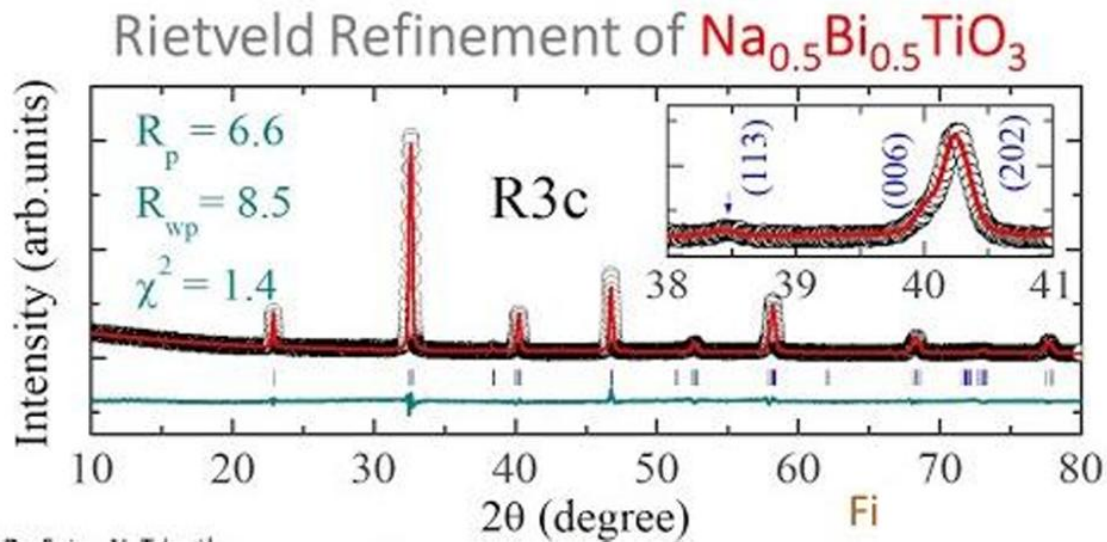
- The graph shows a standard Rietveld refinement plot for the CeO_2 metal oxide sample, comparing experimental data with the calculated crystal structure model.

Case Study 2: Determination of lattice parameters for metal oxide material (ZnO).



- The ZnO refinement plot displays a precise visual overlap between the experimental diffraction data and the calculated structural model.
- This excellent fit is confirmed by the remarkably flat difference line at the base, verifying minimal residual error and highly accurate peak profile modeling.

Case Study 3: Determination of lattice parameters for doped metal oxide material ($\text{Na}_{0.5}\text{Bi}_{0.5}\text{TiO}_3$).



- The graph demonstrates a high-quality structural fit with precise overlap between the observed and calculated diffraction profiles for the rhombohedral .

Conclusion:

- The Rietveld refinement using FullProf successfully determined the lattice parameters of the tested materials.
- The low R_{wp} and χ^2 values confirm good agreement between observed and calculated patterns, validating the reliability of the refinement process.